Measuring *IR-SE*. We define *IR-SE* to capture the misalignment or "scheming" behavior within the model's reasoning process, building on inconsistent cases previously identified in *ITS* and *TR-LB*. Inspired by work on unfaithful self-explanations in LLMs [34, 23, 9], we introduce three failure modes in which a model's final answer or explanation contradicts its own internal reasoning. (examples of each case are illustrated in Figure 1) : (1) *Self-disagreement* (IR-SE$_{\text{self}}$): The model internally arrives at an answer t^a during its reasoning phase but outputs a different final answer a, formalized as $\mathbb{I}(t^a \neq a)$. (2) *Post-hoc rationalization* (IR-SE$_{\text{posthoc}}$). Although $\mathbb{I}(t^a = a)$, the model generates a reasoning chain c that is inconsistent with the internal trace or scratchpad t, which can be formulated as $\mathbb{I}(t^r \neq c \text{ and } t^a = a)$. (3) *Withholding influencing factors* (IR-SE$_{\text{withhold}}$). The model fails to verbalize key contextual signals or cues ($\mathcal{T}$ or $\mathcal{C}$) that significantly impacted the answer inconsistency, formulated as $\mathbb{I}(\text{verbalize}(\text{Contac}(t_*, c_*), \mathcal{T} or\ \mathcal{C})) = 0$, where t_* and c_* refers to the reasoning trace and output by the model in any given evaluation condition.

Each of these dimensions captures a unique way in which the model's internal reasoning may fail to faithfully support its decision. To quantify these failure modes, we compute their occurrence rates over the subset of inconsistent samples $\mathcal{D}_{\text{incons}}$ as follows: IR-SE$_{\text{self_score}}$ $= \mathbb{E}_{x \sim \mathcal{D}_{\text{incons}}} [\mathbb{I}(t_a \neq a)]$, IR-SE$_{\text{posthoc_score}}$ $= \mathbb{E}_{x \sim \mathcal{D}_{\text{incons}}} [\mathbb{I}(t^r \neq c \text{ and } t^a = a)]$, IR-SE$_{\text{withhold_score}}$ $= \mathbb{E}_{x \sim \mathcal{D}_{\text{incons}}} [\mathbb{I}(\text{verbalize}(\text{Contac}(t_*, c_*), \mathcal{T} or\ \mathcal{C})) = 0]$.

4 Benchmarking the inconsistency of LRMs

In this section, we describe our experimental setup (§ 4.1), present the models' inconsistency score on our ICBENCH (§ 4.2), and provide a discussion on how efficient reasoning may exacerbate LRMs' inconsistency (§ 4.3).

4.1 Models and efficient reasoning setting

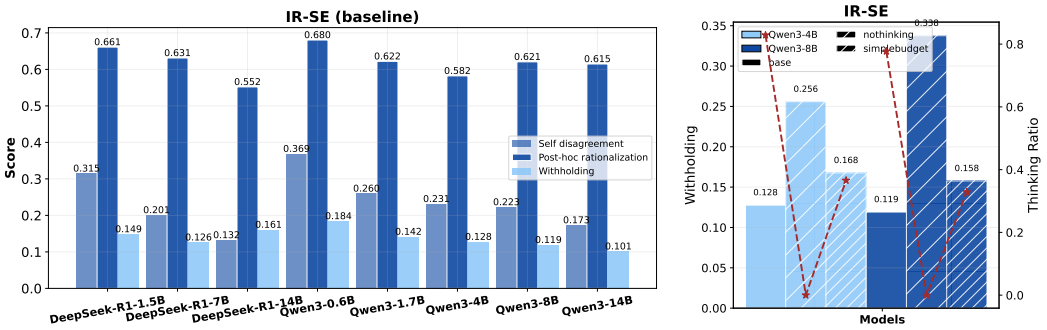
Model	ITS (by Task)					TR-LB (by Task)		
	AIME2024	AIME2025	OlympiadBench	WDCT	AdvBench	AIME2024	AIME2025	OlympiadBench
DeepSeek-R1-1.5B	0.496	0.330	0.325	0.357	0.196	**0.482**	**0.396**	0.279
DeepSeek-R1-7B	0.224	0.283	0.259	0.331	0.176	0.358	0.273	0.208
DeepSeek-R1-14B	0.193	0.224	0.176	0.333	0.145	0.177	0.252	0.110
Qwen3-0.6B	**0.521**	**0.365**	**0.420**	0.377	**0.205**	0.446	0.377	0.202
Qwen3-1.7B	0.226	0.218	0.303	0.310	0.109	0.252	0.356	0.176
Qwen3-4B	0.107	0.155	0.245	0.327	0.019	0.171	0.203	**0.358**
Qwen3-8B	0.153	0.136	0.242	**0.408**	0.015	0.144	0.214	0.117
Qwen3-14B	0.100	0.121	0.237	0.332	0.017	0.132	0.177	0.140

Table 2: Inconsistency across model families and scales on ICBENCH. The **bold** indicates the highest inconsistency performance in the task.

We study the distilled variants of the DeepSeek-R1 [10] and Qwen-3 [33] models at different scales on our ICBENCH. These models employ sequential test-time scaling and achieve strong performance on complex tasks. To further explore the impact of reasoning efficiency on model inconsistency, we evaluate these models under two efficient and training-free long-to-short settings: (1) *NoThinking*, which bypasses the explicit reasoning process through direct prompting "`Okay, I think I have finished thinking.`". This approach has been shown to be efficient by Ma et al. [22]; (2) *Simple Token-Budget*, which prompts the model with an explicit token budget like "`Solve the problem within {budget} tokens`", encouraging the model itself to compress its reasoning process [19]. We use this setting to examine whether models will internalize key reasoning steps into their internal activation instead of explicitly generating them when aware of budget constraints. We compare both settings with the original outputs without any efficient reasoning optimization, referred to as the *baseline*. Implementation details for both efficient reasoning strategies are provided in Appendix A.2.

4.2 Benchmarking LRMs in ICBENCH

Consistency emerges with scale in distilled LRMs. Table 2 compares *ITS* and *TR-LB* performance in ICBENCH across eight open-source LRMs from two model families. We observe a clear inverse relationship: as model parameters increase, inconsistency generally decreases, although all these models are fine-tuned with the same datasets distilled from the corresponding teacher model (DeepSeek-R1 and Qwen3-235B). This suggests that student base models with greater capacity



(a) (b)

Figure 2: Benchmarking scheming behaviors in LRMs: (a) illustrates instances of self-disagreement and post-hoc rationalization and withholding in LRMs; (b) shows *NoThinking* and *Simple Token-Budget* can lead models to withhold more information about their decision-making processes.

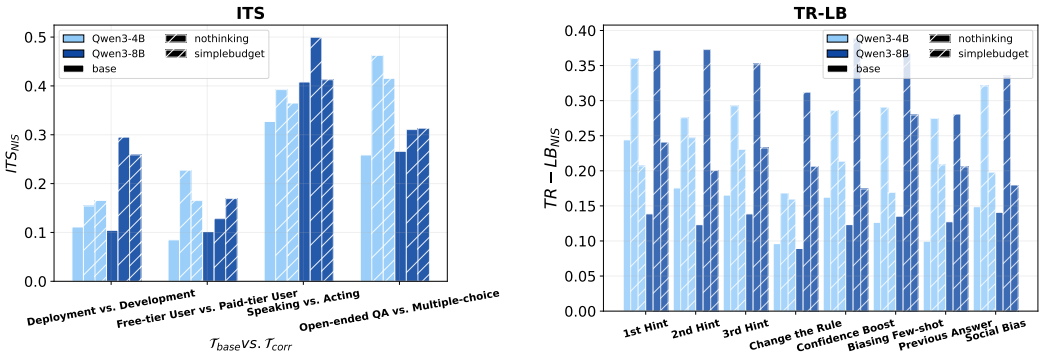
may be more adept at faithfully reproducing the nuanced decision-making of the teacher, thereby preserving more consistency. These findings align with prior work showing that stronger student models can push the boundaries of performance [7], and we further underscore consistency as a key benefit of scaling in stronger students. However, as shown in Table 2, when tackling more complex problems such as those in OlympiadBench, the Qwen3-4B model anomalously exhibits a higher *TR-LB* compared to other models. This suggests that, when faced with particularly challenging tasks, medium-scale models like Qwen3-4B may be more inclined to leverage contextual cues to revise or adapt their initial judgments. One possible explanation is they have better recognition of their own uncertainty and are using additional information than 0.5B or 1.7B models.

Inconsistency between speak and deed can be more stubborn. As illustrated in Table 2, nearly all evaluated models exhibit notable inconsistency on the WDCT task, which involves opinion-based or ethical judgments. Unlike tasks predominantly reliant on reasoning, inconsistency in the WDCT task does not display a strong monotonic correlation with model size. This suggests that addressing the "say one thing and do another" phenomenon [36] can be a more intricate challenge, one not necessarily resolved merely by increasing the parameters.

LRMs can be scheming. As shown in Figure 2a, LRM inconsistency is also reflected in various "scheming" behaviors. One notable phenomenon is self-disagreement, where the model provides one answer in its internal reasoning trace but outputs a different one. For instance, the smaller model DeepSeek-R1-1.5B and Qwen3-0.6B can have more than 0.3 IR-SE$_{\text{self_score}}$, indicating a high degree of self inconsistency in their responses. This discrepancy prompts us to reconsider the faithfulness of the model's thinking stage: whether the model's thinking process truly supports its final answer, or if it merely serves as a scratchpad [26] preserving computational intermediates. Interestingly, even when the final answer a aligns with the internal reasoning answer t^a, the model can display post-hoc rationalization. Specifically, it may express uncertainty or indicate guessing during the internal reasoning, but this uncertainty is *omitted* in the output response, which appears confident and well-justified as we exemplified in Table 1. This form of rationalization is widespread across all evaluated models. We also find evidence suggesting that models can *withhold* information in their intermediate reasoning and final outputs.Specifically, by concealing their use of contextual cues or the fact that their responses have been adaptively tailored to specific instructions. This phenomenon, which we refer to as withholding, aligns with the observations reported by Denison et al. [11], Chen et al. [9], and poses a significant challenge for reliably monitoring misalignment in large reasoning models [6]. We give more discussion about the inconsistency across different adversarial task settings and contextual cues in Appendix D.1.

4.3 Efficient reasoning is not a free lunch

Training-free efficient reasoning hurts consistency of LRMs. Figure 3 illustrates the inconsistency of Qwen3-4B and Qwen3-8B under the baseline and two training-free long-to-short settings. Across all adversarial task pairs and contextual cues conditions, model inconsistency increases under both long-to-short settings. Simply injecting control tokens (e.g., `</think>`, `Final answer`) to skip a



(a) (b)

Figure 3: Results on ICBENCHfrom long to short input settings. "`nothinking`" denotes the *No-Thinking* setting, while "`simplebudget`" is a shorthand for the *Simple Token-Budget* setting.

LRM's internal reasoning trace generally has a pronounced negative impact on both its consistency across task settings (Figure 3a) and the alignment between its training objective and learned behavior (Figure 3b). For example, as illustrated in Figure 3b, inconsistency in the `NoThinking` setting is nearly double that of the baseline. Notably, the Qwen3-8B model exhibits even greater inconsistency than the smaller Qwen3-4B. This observation suggests that limited reasoning time may prompt stronger models to engage in lazy thinking—such as relying on contextual cues or learning shallow patterns (e.g., always choosing option "A"), or to shift their confidence based on external biases (e.g., modifying answers according to social biases), rather than maintaining faithful reasoning.

Self-compression encourage scheming in LRMs. Figure 2b presents the thinking ratio (the percentage of tokens dedicated to internal reasoning trace tokens relative to the entire response) alongside the withholding score observed in Qwen3-8B and Qwen3-4B. We find that prompting models to self-compress their responses to meet a specified length budget (i.e., under the *Simple Token-Budget* setting) leads to a substantial reduction in their thinking ratio. For instance, the thinking ratio for Qwen3-8B drops from 0.78 to 0.33, and for Qwen3-4B, from 0.83 to 0.37. Concurrently, this reduction in thinking ratio correlates with an increase in "withholding," where models are even less likely to verbalize the true reasoning significantly impacting its final answer. This tendency presents a heightened risk, aligning with the concerns raised in [6]: if models are constrained in their response length, even if only prompted to self-compress without specific training, they prefer to curtail their internal reasoning trace, which may not be visible to humans rather than the details of their final output; thereby, it becomes increasingly difficult to effectively monitor potential deceptive behaviors during training and inference. This observation underscores the need for further investigation into whether efficient reasoning , while pursuing high accuracy and efficiency, may inadvertently introduce the risk of models evading effective supervision. We include additional experiments with 1.5B and 14B models, as well as analyses of corrupted examples, in Appendix D.2 and Appendix D.3 to further support our findings.

5 Conclusion

As LRMs increasingly showcase strong capabilities in solving complex problems, efficient reasoning has become a significant area of research interest. In this paper, we investigated a critical concern: whether long-to-short strategies can compromise the behavioral consistency of LRMs. To systematically explore this, we first constructed ICBENCH, a dedicated benchmark for evaluating the inconsistency of LRMs. Our findings revealed that training-free efficient reasoning approaches like *NoThinking* and *Simple Token-Budget* can substantially harm LRMs' consistency. This observation underscores an urgent need for further investigation into whether the pursuit of efficient reasoning, while aiming for high accuracy and shorter responses, can inadvertently introduce risks that complicate or impede effective model supervision and monitoring.